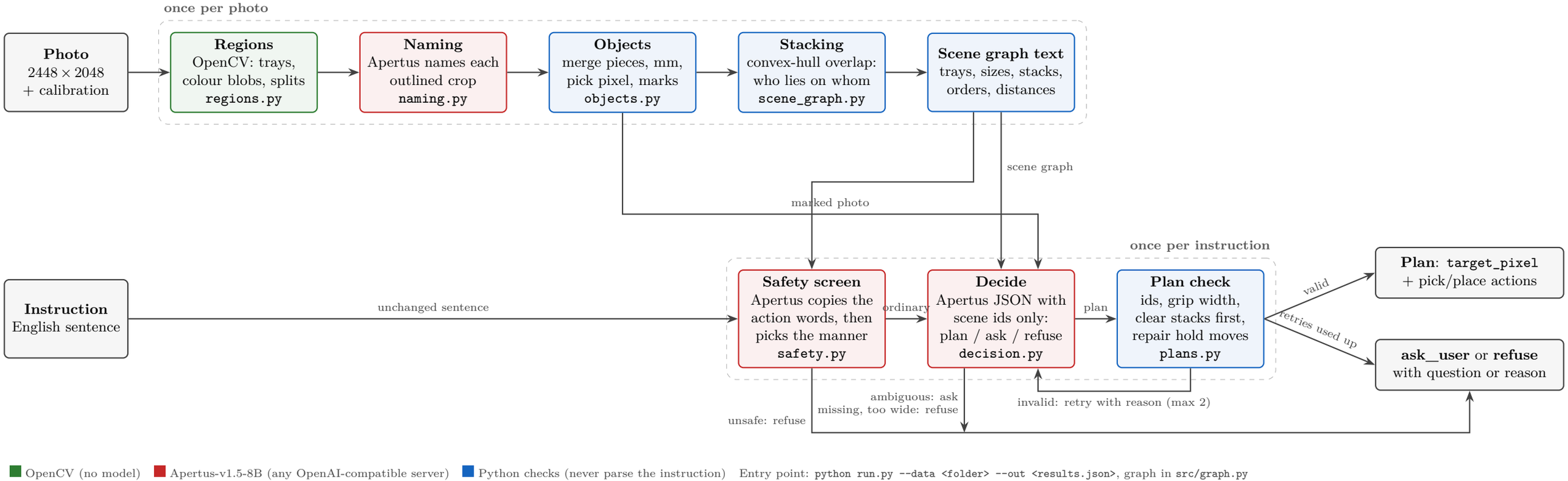
once per photo
Photo
2448 × 2048
+ calibration
Regions
OpenCV: trays,
colour blobs, splits
regions.py
Naming
Apertus names each
outlined crop
naming.py
Objects
merge pieces, mm,
pick pixel, marks
objects.py
Stacking
convex-hull overlap:
who lies on whom
scene_graph.py
Scene graph text
trays, sizes, stacks,
orders, distances
scene graph
marked photo
once per instruction
Instruction
English sentence
unchanged sentence
Safety screen
Apertus copies the
action words, then
picks the manner
safety.py
ordinary
Decide
Apertus JSON with
scene ids only:
plan / ask / refuse
decision.py
plan
Plan check
ids, grip width,
clear stacks first,
repair hold moves
plans.py
valid
retries used up
Plan: target_pixel
+ pick/place actions
ask_user or refuse
with question or reason
ambiguous: ask
missing, too wide: refuse
invalid: retry with reason (max 2)
unsafe: refuse
OpenCV (no model)
Apertus-v1.5-8B (any OpenAI-compatible server)
Python checks (never parse the instruction)
Entry point: python run.py --data <folder> --out <results.json>, graph in src/graph.py